

Method of separation of variables.

1) Solve by method of separation of variables

$$\frac{\partial y}{\partial x} = 2 \frac{\partial y}{\partial t} + u ; \text{ where } u(x,0) = 6e^{-3x}$$

Sol Let the soln of this eqn be

$$u(x,t) = X(x) \cdot T(t).$$

$$\Rightarrow \frac{\partial y}{\partial x} = X' \cdot T, \quad \frac{\partial y}{\partial t} = X T'$$

$$\Rightarrow X' T = 2 X T' + X T$$

$$\Rightarrow (X' - X) T = 2 X T'$$

$$\Rightarrow \frac{X' - X}{X} = \frac{2 T'}{T} = K \text{ (say)}$$

Solving $\frac{X' - X}{X} = K$

$$\Rightarrow \frac{X'}{X} - 1 = K \Rightarrow \frac{X'}{X} = K + 1$$

ie $\left(\frac{dx}{x} = K+1\right)$ on integrating

$$\Rightarrow \int \frac{X'}{X} = \int (K+1) dx \Rightarrow \int \frac{dx}{x} = \int (K+1) dx$$

$$\log x = (K+1)x + \log C_1$$

$$\Rightarrow \log x = \log C_1 + (K+1)x$$

$$\Rightarrow \log \frac{x}{C_1} = (K+1)x$$

$$\Rightarrow \frac{x}{C_1} = e^{(K+1)x} \Rightarrow x = C_1 e^{(K+1)x}$$

$$\Rightarrow \frac{x}{C_1} = e$$

$$\Rightarrow \text{solving } \frac{T'}{T} = \frac{k}{2} \Rightarrow \frac{dT}{T} = \frac{k}{2}$$

$$\text{on integrating } \log T = \frac{k}{2} \cdot t + \log C_2$$

$$\log T - \log C_2 = \frac{kt}{2}$$

$$\log \frac{T}{C_2} = \frac{kt}{2}$$

$$\frac{T}{C_2} = e^{\frac{kt}{2}}$$

$$T = C_2 e^{\frac{kt}{2}}$$

$$\therefore u(x, t) = XT = C_1 e^{(k+1)x} \cdot C_2 e^{\frac{kt}{2}}$$

$$= C_1 C_2 e^{(k+1)x + \frac{kt}{2}}$$

$$\text{Now using } u(x, 0) = 6e^{-3x}$$

$$\text{when } t=0, u = 6e^{-3x}$$

$$\Rightarrow 6e^{-3x} = C_1 C_2 e^{(k+1)x + 0}$$

$$\Rightarrow 6e^{-3x} = A e^{(k+1)x} \quad (C_1 C_2 = A)$$

$$\Rightarrow \text{So } A = 6, \quad k+1 = -3$$

$$k = -3 - 1 \Rightarrow k = -4$$

$$u(x, t) = 6 e^{(-4+1)x - \frac{4t}{2}}$$

$$= 6 e^{-3x - 2t}$$

$$u = 6 e^{-3x - 2t}$$

$$u = 6 e^{-(3x+2t)} \text{ Ans.}$$

② Solve by the method of separation of variables

$$\frac{\partial u}{\partial x} = 4 \frac{\partial u}{\partial y}, \text{ given that } u(0, y) = 8e^{-3y}$$

Sol Given $\frac{\partial u}{\partial x} = 4 \frac{\partial u}{\partial y}$

Let the soln of this eqn

$$u(x, y) = X(x) \cdot Y(y)$$

$$\Rightarrow \frac{\partial u}{\partial x} = X' Y, \quad \frac{\partial u}{\partial y} = X Y'$$

$$\Rightarrow X' Y = 4 X Y'$$

$$\Rightarrow \frac{X'}{X} = \frac{4 Y'}{Y} = K$$

$$\text{Solving } \frac{X'}{X} = K \Rightarrow \int \frac{dX}{X} = K \int dx$$

$$\Rightarrow \text{on integrating } \log X = Kx + \log C_1$$

$$\Rightarrow \log X - \log C_1 = Kx$$

$$\Rightarrow \log \frac{X}{C_1} = Kx$$

$$\Rightarrow \frac{X}{C_1} = e^{Kx} \Rightarrow X = C_1 e^{Kx}$$

$$\Rightarrow \text{Solving } \frac{4 Y'}{Y} = K \Rightarrow \frac{Y'}{Y} = \frac{K}{4}$$

$$\Rightarrow \frac{dY}{Y} = \frac{K}{4}$$

$$\text{on integrating } \log Y = \frac{K}{4} y + \log C_2$$

$$\log \frac{Y}{C_2} = \frac{K}{4} y \Rightarrow \frac{Y}{C_2} = e^{\frac{K}{4} y}$$

$$\frac{Y}{C_2} = e^{\frac{Ky}{4}}$$

$$Y = C_2 e^{\frac{Ky}{4}}$$

$$\begin{aligned} \therefore u(x, y) &= XY \\ &= C_1 e^{Kx} \cdot C_2 e^{\frac{Ky}{4}} \\ &= C_1 C_2 e^{\frac{Kx + Ky}{4}} \end{aligned}$$

$$u(0, y) = 8e^{-3y}$$

$$\text{when } x=0 \Rightarrow u = 8e^{-3y}$$

$$\Rightarrow 8e^{-3y} = C_1 C_2 e^{0 + \frac{Ky}{4}}$$

(let $C_1 C_2 = A$)

$$\Rightarrow 8e^{-3y} = Ae^{\frac{ky}{4}}$$

$$A = 8, \quad \frac{k}{4} = -3 \Rightarrow k = -12$$

$$\text{so } u = Ae^{kx + \frac{ky}{4}}$$

$$\Rightarrow u = 8e^{-12x - \frac{12y}{4}}$$

$$\Rightarrow u = 8e^{-12x - 3y}$$

$$\Rightarrow u = e^{-3(4x+y)} \text{ Ans.}$$

3) Solve by method of separation of variables

$$4 \frac{\partial y}{\partial x} + \frac{\partial y}{\partial y} = 3u, \text{ given that -}$$

$$u = 3e^{-y} - e^{-5y} \text{ when } x = 0.$$

Sol let the solⁿ be

$$u(x, y) = X(x) \cdot Y(y)$$

$$\Rightarrow 4 \cancel{xy'} = 4x'y + xy' = 3xy$$

$$\Rightarrow 4x'y \neq (3y - y')x$$

$$\Rightarrow \frac{4x'}{y} = \frac{3y - y'}{y} = k$$

$$\text{Solving } \frac{4x'}{x} = k \Rightarrow \frac{x'}{x} = \frac{k}{4}$$

on integrating $\log x = \frac{k}{4} x + \log c_1$

$$\log \frac{x}{c_1} = \frac{k}{4} x$$

$$\frac{x}{c_1} = e^{\frac{kx}{4}} \Rightarrow x = c_1 e^{\frac{kx}{4}}$$

Solving $\frac{3y - y'}{y} = k$

$$\Rightarrow 3 - \frac{y'}{y} = k$$

$$\Rightarrow 3 - k = \frac{y'}{y}$$

$\Rightarrow$ on integrating

$$\Rightarrow \log y = (3 - k)y + \log c_2$$

$$\Rightarrow \log \frac{y}{c_2} = (3 - k)y$$

$$\Rightarrow \frac{y}{c_2} = e^{(3 - k)y}$$

$$\Rightarrow y = c_2 e^{(3 - k)y}$$

Now $u = x(n) \cdot y(y)$

$$u = c_1 e^{\frac{kx}{4}} \cdot c_2 e^{(3 - k)y}$$

$$u = c_1 c_2 e^{\frac{kx + (3 - k)y}{4}}$$

$$c_1 c_2 = A$$

$$\Rightarrow u = A e^{\frac{kx}{4} + \frac{(3-k)y}{4}}$$

when $x=0 \Rightarrow u = 3e^{-y} - e^{-5y}$

$$\Rightarrow 3e^{-y} - e^{-5y} = A e^{(3-k)y}$$

$\Rightarrow$ on comparing

$$A e^{(3-k)y} = 3e^{-y}$$

$$A = 3, \quad 3 - k = -1 \Rightarrow k = 4$$

$$\rightarrow \text{so } u = A e \left[\cancel{\frac{4x}{4}} + (3-4)y \right]$$

$$\rightarrow u = 3 e^{x-y} \text{ ——— ①}$$

Now comparing,

$$A e^{(3-k)y} = -e^{-5y}$$

$$A = -1, \quad 3-k = -5 \Rightarrow 3+5=k$$

$$\Rightarrow k = 8$$

$$\Rightarrow u = -e^{\left[\frac{8x}{4} + (3-2)y\right]}$$

$$u = -e^{2x-5y} \quad \text{--- (2)}$$

from (1) & (2)

$$u(x, y) = 3e^{x-y} - e^{2x-5y}$$

which is the required soln.

4) Solve

$$4 \frac{\partial y}{\partial u} + \frac{\partial y}{\partial y} = 3u \quad \& \quad u = e^{-5y} \text{ when}$$

$$(\text{Ans} \rightarrow u = e^{2x-5y})$$

$$x=0.$$

$$5) \quad 3 \frac{\partial y}{\partial x} + 2 \frac{\partial y}{\partial y} = 0, \quad u(x, 0) = 4e^{-x}$$

$$(\text{Ans} = \cancel{u = 4e^{\frac{-x+3}{2y}}} \quad u = 4e^{-x + \frac{3}{2}y}).$$